BrucellaID
Brucella species identification and MLST from reads

Quality assurance report

4 sample(s): 4 ok, 0 with a warning, 0 failed. BrucellaID 1.2.1, run on 2026-11-02T01:11:53-05:00.

This is an example report: a real run of the four public read sets of the BrucellaID tutorial, produced by scripts/make_example_report.py, which leaves out the computer it ran on. The user account, the host name, the operating system and the working directory read [removed], and the input and reference files are named without the folder they were in. Everything else is as BrucellaID wrote it. A report of your own names them all, which is what makes it worth signing.

1 Summary

Run started	2026-11-02T01:11:53-05:00
Run finished	2026-11-02T01:12:26-05:00
Operator	M-O Duceppe
Command	python3 scripts/make_example_report.py reads

SAMPLE	SPECIES	FROM	DECISION	ST	ALLELE PROFILE	CLOSEST REFERENCE	STATUS
Brucella abortus · ST 1 (1 sample(s))							
ERR3454415	Brucella abortus	MASH+ST	agree	1	2-1-1-2-1-3-1-1-1	NC_006932-NC_006933.fasta	**ok**
Brucella canis · ST 20 (1 sample(s))							
ERR2136546	Brucella canis	MASH+ST	arbitrated	20	1-6-4-1-5-3-5-4-6	NC_010103-NC_010104.fasta	**ok**
Brucella melitensis · ST 11 (1 sample(s))							
SRR19520329	Brucella melitensis	MASH+ST	agree	11	3-2-3-2-1-5-3-2-10	NZ_CP007760-NZ_CP007761.fasta	**ok**
Brucella suis · ST 15 (1 sample(s))							
SRR8550484	Brucella suis	MASH+ST	agree	15	1-2-7-1-3-3-5-3-2	NC_010169-NC_010167.fasta	**ok**

REVIEWED BY

Name Date Signature

2 Notes, warnings and errors

SAMPLE	STATUS	WHAT IT SAYS
ERR2136546	ok	nearly as close: B-REF-BS4-40.fasta (Brucella suis bv. 4, 768/1000 hashes)
ERR2136546	ok	MASH cannot separate Brucella canis and Brucella suis: ST 20 is Brucella canis in 70 of 70 isolate(s), which agrees with the closest reference

3 Samples

ERR3454415 ok Brucella abortus ST 1

IDENTIFICATION: THE MASH DISTANCE

Closest reference	NC_006932-NC_006933.fasta
What it is	Brucella abortus bv. 1 str. 9-941
NCBI accessions	NC_006932.1, NC_006933.1
MASH distance	0.000656
Estimated identity	99.934%
Shared min-hashes	973/1000 (at least 150 required)
K-mer copies required	3 (from the estimated 70x coverage)

THE SPECIES CALL

Species reported	Brucella abortus
Decided by	MASH+ST
Species within reach	Brucella abortus
Sequence type	1
What that ST is	ST 1 is Brucella abortus in 463 of 466 isolate(s) (the rest: Brucella melitensis:3)
Biovars of its isolates	1:86, 2:16, 3:1, 4:7

agree The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 11 reference(s) are further away; brucellaid_mash.tsv has every distance.

REFERENCE GENOME	SPECIES	STRAIN	DISTANCE	IDENTITY %	SHARED HASHES
NC_006932-NC_006933.fasta	Brucella abortus	9-941	0.000656	99.934	973/1000
NZ_CP007682-NZ_CP007683.fasta	Brucella abortus	BER	0.001786	99.821	929/1000
NZ_CP007760-NZ_CP007761.fasta	Brucella melitensis	Ether	0.002771	99.723	893/1000
NC_012441-NC_012442.fasta	Brucella melitensis	ATCC 23457	0.002799	99.720	892/1000
NZ_CP018508-NZ_CP018509.fasta	Brucella melitensis	BwIM_SOM_36b	0.002799	99.720	892/1000

MLST: THE SEQUENCE TYPE

Sequence type	1
Allele profile	2-1-1-2-1-3-1-1-1
Every locus called	yes
Reads in the input	1,175,082
Reads matching an allele	2,948

INPUT FILES

FILE	SIZE	MODIFIED
ERR3454415_1.fastq.gz	87.4 MB	2026-11-01 23:41:18 EST
ERR3454415_2.fastq.gz	83.5 MB	2026-11-01 23:47:42 EST

```
ERR3454415_1.fastq.gz
MD5 8eaddc2c8044657712ec586ba5482c70
ERR3454415_2.fastq.gz
MD5 b2706df339382a63f38bba34222b4053
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

LOCUS	ALLELE	READS	RUNNER-UP (ITS READS)	AHEAD	RATIO	NOTE
gap	2	432	gap_6 (300)	132	1.44	
aroA	1	322	aroA_30 (210)	112	1.53	
glk	1	222	glk_21 (158)	64	1.41	
dnaK	2	340	dnaK_26 (222)	118	1.53	
gyrB	1	334	gyrB_22 (224)	110	1.49	
trpE	3	320	trpE_4 (242)	78	1.32	
cobQ	1	308	cobQ_12 (196)	112	1.57	
int_hyp	1	368	int_hyp_4 (256)	112	1.44	
omp25	1	292	omp25_12 (186)	106	1.57	

ERR2136546 ok Brucella canis ST 20

IDENTIFICATION: THE MASH DISTANCE

Closest reference	NC_010103-NC_010104.fasta
What it is	Brucella canis str. ATCC 23365
NCBI accessions	NC_010103.1, NC_010104.1
MASH distance	0.006524
Estimated identity	99.348%
Shared min-hashes	773/1000 (at least 150 required)
K-mer copies required	2 (from the estimated 47x coverage)

THE SPECIES CALL

Species reported	Brucella canis
Decided by	MASH+ST
Species within reach	Brucella canis, Brucella suis
Sequence type	20
What that ST is	ST 20 is Brucella canis in 70 of 70 isolate(s)

arbitrated A reference of a second species was near enough to the closest one that the distance did not really choose. The sequence type chose between the species it left within reach.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 11 reference(s) are further away; brucellaid_mash.tsv has every distance.

REFERENCE GENOME	SPECIES	STRAIN	DISTANCE	IDENTITY %	SHARED HASHES
NC_010103-NC_010104.fasta	Brucella canis	ATCC 23365	0.006524	99.348	773/1000
B-REF-BS4-40.fasta	Brucella suis	40	0.006698	99.330	768/1000
NC_017251-NC_017250.fasta	Brucella suis	1330	0.007196	99.280	754/1000
NZ_CP007719-NZ_CP007718.fasta	Brucella suis	686	0.007268	99.273	752/1000
NC_013119_NC_013118.fasta	Brucella microti	CCM 4915	0.008080	99.192	730/1000

MLST: THE SEQUENCE TYPE

Sequence type	20
Allele profile	1-6-4-1-5-3-5-4-6
Every locus called	yes
Reads in the input	625,576
Reads matching an allele	1,824

INPUT FILES

FILE	SIZE	MODIFIED
ERR2136546_1.fastq.gz	57.0 MB	2026-11-01 23:50:46 EST
ERR2136546_2.fastq.gz	63.9 MB	2026-11-01 23:43:00 EST

```
ERR2136546_1.fastq.gz
MD5 7573e407362c1458a464a4d92fec2db4
ERR2136546_2.fastq.gz
MD5 a83a9dceaee29371219b98f0a2f1dc39
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

LOCUS	ALLELE	READS	RUNNER-UP (ITS READS)	AHEAD	RATIO	NOTE
gap	1	232	gap_26 (170)	62	1.36	
aroA	6	194	aroA_31 (80)	114	2.42	
glk	4	148	glk_2 (94)	54	1.57	
dnaK	1	244	dnaK_2 (162)	82	1.51	
gyrB	5	224	gyrB_4 (110)	114	2.04	
trpE	3	178	trpE_5 (114)	64	1.56	
cobQ	5	172	cobQ_30 (108)	64	1.59	
int_hyp	4	234	int_hyp_1 (180)	54	1.30	
omp25	6	172	omp25_2 (98)	74	1.76	

NOTES

- nearly as close: B-REF-BS4-40.fasta (Brucella suis bv. 4, 768/1000 hashes)
- MASH cannot separate Brucella canis and Brucella suis: ST 20 is Brucella canis in 70 of 70 isolate(s), which agrees with the closest reference

SRR19520329 ok Brucella melitensis ST 11

IDENTIFICATION: THE MASH DISTANCE

Closest reference	NZ_CP007760-NZ_CP007761.fasta
What it is	Brucella melitensis bv. 3 str. Ether
NCBI accessions	NZ_CP007760.1, NZ_CP007761.1
MASH distance	0.000582
Estimated identity	99.942%
Shared min-hashes	976/1000 (at least 150 required)
K-mer copies required	3 (from the estimated 53x coverage)

THE SPECIES CALL

Species reported	Brucella melitensis
Decided by	MASH+ST
Species within reach	Brucella melitensis
Sequence type	11
What that ST is	ST 11 is Brucella melitensis in 546 of 546 isolate(s)
Biovars of its isolates	1:1, 2:7, 3:12

agree The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 11 reference(s) are further away; brucellaid_mash.tsv has every distance.

REFERENCE GENOME	SPECIES	STRAIN	DISTANCE	IDENTITY %	SHARED HASHES
NZ_CP007760-NZ_CP007761.fasta	Brucella melitensis	Ether	0.000582	99.942	976/1000
NC_012441-NC_012442.fasta	Brucella melitensis	ATCC 23457	0.001263	99.874	949/1000
NZ_CP018508-NZ_CP018509.fasta	Brucella melitensis	BwIM_SOM_36b	0.001263	99.874	949/1000
NC_003317-NC_003318.fasta	Brucella melitensis	16M	0.001627	99.837	935/1000
NC_013119_NC_013118.fasta	Brucella microti	CCM 4915	0.002658	99.734	897/1000

MLST: THE SEQUENCE TYPE

Sequence type	11
Allele profile	3-2-3-2-1-5-3-2-10
Every locus called	yes
Reads in the input	1,136,924
Reads matching an allele	2,824

INPUT FILES

FILE	SIZE	MODIFIED
SRR19520329_1.fastq.gz	35.9 MB	2026-11-01 23:46:06 EST
SRR19520329_2.fastq.gz	40.6 MB	2026-11-01 23:46:59 EST

```
SRR19520329_1.fastq.gz
MD5 12edf200445bafd03d033fac3d98118c
SRR19520329_2.fastq.gz
MD5 a81d3abf43d7a2849c9062061cd8603a
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

LOCUS	ALLELE	READS	RUNNER-UP (ITS READS)	AHEAD	RATIO	NOTE
gap	3	350	gap_34 (250)	100	1.40	
aroA	2	324	aroA_1 (256)	68	1.27	
glk	3	282	glk_26 (184)	98	1.53	
dnaK	2	280	dnaK_6 (204)	76	1.37	
gyrB	1	374	gyrB_2 (266)	108	1.41	
trpE	5	348	trpE_31 (242)	106	1.44	
cobQ	3	260	cobQ_2 (194)	66	1.34	
int_hyp	2	328	int_hyp_20 (222)	106	1.48	
omp25	10	270	omp25_8 (182)	88	1.48	

SRR8550484 ok Brucella suis ST 15

IDENTIFICATION: THE MASH DISTANCE

Closest reference	NC_010169-NC_010167.fasta
What it is	Brucella suis bv. 2 str. ATCC 23445
NCBI accessions	NC_010169.1, NC_010167.1
MASH distance	0.006455
Estimated identity	99.355%
Shared min-hashes	775/1000 (at least 150 required)
K-mer copies required	3 (from the estimated 69x coverage)

THE SPECIES CALL

Species reported	Brucella suis
Decided by	MASH+ST
Species within reach	Brucella suis
Sequence type	15
What that ST is	ST 15 is Brucella suis in 99 of 103 isolate(s) (the rest: Brucella melitensis:2, Brucella ovis:1, Brucella sp.:1)
Biovars of its isolates	2:60

agree The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 11 reference(s) are further away; brucellaid_mash.tsv has every distance.

REFERENCE GENOME	SPECIES	STRAIN	DISTANCE	IDENTITY %	SHARED HASHES
NC_010169-NC_010167.fasta	Brucella suis	ATCC 23445	0.006455	99.355	775/1000
NC_017251-NC_017250.fasta	Brucella suis	1330	0.008118	99.188	729/1000
NC_013119_NC_013118.fasta	Brucella microti	CCM 4915	0.008308	99.169	724/1000
B-REF-BS4-40.fasta	Brucella suis	40	0.008346	99.165	723/1000
NC_010103-NC_010104.fasta	Brucella canis	ATCC 23365	0.008384	99.162	722/1000

MLST: THE SEQUENCE TYPE

Sequence type	15
Allele profile	1-2-7-1-3-3-5-3-2
Every locus called	yes
Reads in the input	1,693,672
Reads matching an allele	3,204

INPUT FILES

FILE	SIZE	MODIFIED
SRR8550484_1.fastq.gz	71.6 MB	2026-11-01 23:48:07 EST
SRR8550484_2.fastq.gz	72.4 MB	2026-11-01 23:50:36 EST

```
SRR8550484_1.fastq.gz
MD5 68833febebe273fadd9dc6ee0e95ed16
SRR8550484_2.fastq.gz
MD5 9b8438792c71722cd7fb4fb0d18e414c
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

LOCUS	ALLELE	READS	RUNNER-UP (ITS READS)	AHEAD	RATIO	NOTE
gap	1	508	gap_14 (412)	96	1.23	
aroA	2	284	aroA_16 (232)	52	1.22	
glk	7	282	glk_4 (202)	80	1.40	
dnaK	1	302	dnaK_27 (224)	78	1.35	
gyrB	3	384	gyrB_1 (290)	94	1.32	
trpE	3	346	trpE_5 (296)	50	1.17	
cobQ	5	294	cobQ_4 (220)	74	1.34	
int_hyp	3	444	int_hyp_1 (272)	172	1.63	
omp25	2	330	omp25_21 (250)	80	1.32	

4 Definitions and methods

The references listed for each sample

A sample block lists the five closest reference genomes, and any other near enough to the closest that the distance does not separate them, which is the evidence the species call rests on. brucellaid_mash.tsv holds the distance to every reference of the sketch.

The species, from a MASH distance

The reads of a sample are reduced to the 1000 smallest hashes of their 21-mers, ignoring the k-mers seen fewer times than the copy threshold derived from the coverage, and that sketch is compared with the sketch of each reference genome. The distance is an estimate of the fraction of positions that differ, so 0.000024 is about 99.998% identity. The closest reference names the species, provided the reads share at least the required number of its 1000 min-hashes; below that the sample is reported as ND, because the closest of sixteen genomes is not an answer.

Why there is no biovar in this report

BrucellaID does not report a biovar, because it cannot measure one. Three things stand in the way. The sketch holds a reference for B. abortus biovars 1 and 3 only, so 43% of the biovar-labelled B. abortus isolates of the PubMLST isolate database are of a biovar it has no reference for. The distance cannot separate the biovars it does hold: the two B. melitensis biovar 1 references are 0.000756 apart, which is further apart than the biovar 2 and biovar 3 references are from each other. And a biovar is a phenotype that genome clustering does not track -- the best any rule from sequence type to biovar could manage over the isolate database is 66% for B. abortus and 63% for B. melitensis, because one ST spans several biovars. The biovars recorded for the ST are shown above for what they are worth. For a biovar, use classical biotyping, or brucetyper, which reads the Bruce-ladder and Suis-ladder assays in silico.

The sequence type, from the alleles

The same reads are given to Seal with the alleles of the nine loci as references, counting each read for every allele it shares the highest number of 25-mers with and for no other. The allele the sample carries is therefore counted for every read of its locus, and a neighbouring allele only for the reads that fall where the two agree, so the allele with the most reads wins. No mismatch is allowed in a k-mer, because the alleles of a locus differ by as little as one base. Two alleles with the same count are reported as a tie rather than separated by a coin toss, and a locus below the minimum read count is not called.

What the ST values mean

A number is a profile that is in the table. "new" means all nine alleles were called but the combination is not in the table, which is either a genuinely new ST or a novel allele called as its nearest neighbour. "ND" means at least one locus had no call; the note says which.

How the species is decided when two are equally close

Some Brucella species are closer to each other than the variation within either of them: in the sketch that ships with BrucellaID, B. canis and B. suis bv. 4 are 0.00024 apart, the closest pair of the sixteen. Where a reference of a second species is within the close-match margin of the closest one, the distance has not really chosen, and the sequence type chooses among those species, because the two share no ST. An ST is only used when enough of its isolates agree on a species, and an ST that contradicts a species the distance picked cleanly is reported as a conflict and not obeyed.

What this report does not establish

That the sample is what the label on the tube says, that the library is not a mixture of two isolates, or that the reference data is current. It establishes what the software was given, what it measured, and what it concluded from those measurements.

5 Run information

Operator	M-O Duceppe
User account	[removed]
Computer	[removed]
Operating system	[removed]
Working directory	[removed]
Started	2026-11-02T01:11:53-05:00
Finished	2026-11-02T01:12:26-05:00
Command	python3 scripts/make_example_report.py reads

Software

PROGRAM	VERSION
brucellaid	1.2.1
Python	3.12.16
MASH	2.3
BBTools	40.05
Java	openjdk version "25.0.4.1" 2026-08-18

Parameters

PARAMETER	VALUE
Threads	10
Parallel samples	2
MLST	True
Input checksums	MD5

Reference data

DATA	WHAT IT HOLDS	SOURCE	MD5
MASH sketch	16 reference genome(s), k = 21, 1000 hashes	the package's	44117b188407e97c5caff376f6e3b4eb
Reference table	16 genome(s) with their species, biovar, strain and accessions	the package's	d221a221f85377746fbbbb88906984dc
MLST alleles	309 allele(s) of the 9 loci	the package's	95fcf5e43f4cbf7ebe840c8e54bfbf46
MLST profiles	122 sequence type(s)	the package's	dd4bc06867bd58ce2a8f362755aa79a7
ST to species	118 sequence type(s) with a species	the package's	3a6751b8a1ab1273f3bc0e19e08efb7f

```
.../brucellaid/data/references.msh
.../brucellaid/data/references.tsv
.../brucellaid/data/mlst_alleles.fasta
.../brucellaid/data/mlst_profiles.tsv
.../brucellaid/data/mlst_st_species.tsv
```

References

1. Whatmore AM, Perrett LL, MacMillan AP. Characterisation of the genetic diversity of Brucella by multilocus sequencing. BMC Microbiology 2007;7:34. doi:10.1186/1471-2180-7-34
2. Ondov BD, Treangen TJ, Melsted P, et al. Mash: fast genome and metagenome distance estimation using MinHash. Genome Biology 2016;17:132. doi:10.1186/s13059-016-0997-x
3. Jolley KA, Bray JE, Maiden MCJ. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. Wellcome Open Research 2018;3:124. doi:10.12688/wellcomeopenres.14826.1
4. Bushnell B. BBTools. https://sourceforge.net/projects/bbmap/